A tidal screen with one constant fixed by the Oort cloud

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Abstract

A tidal screen adds one constant, C, to a two-field Newtonian theory. The constant drops out of the rotation-curve law and remains in the shell stress. At $C = 0$ the shell pushes new comets below the observed Oort peak. The value used here is the centre of the window that puts that peak back, $C = 0.579125$ at a Galactic field $1.9 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. At that value the Cassini quadrupole is unchanged, a clean Gaia wide-binary cut prefers the model to Newton, 171 SPARC galaxies are unchanged at the level of the fit, and four dwarfs take stellar mass-to-light ratios. The same clean cut was repeated at the two edges of the window. All three pass. A radial integration of comet orbits agrees with the energy proxy to $1.3 \times 10^{-6}\,\mathrm{au^{-1}}$. The non-relativistic action that produces the field equation is written down. The constant C is not fixed by that action. A relativistic completion in which the tide invariant is dynamical, a full comet population in the sense of Vokrouhlický, Nesvorný and Tremaine, and a cosmological sector are not in this paper.

1 Field equation

The Newtonian potential is kept. A second potential obeys

$$Q_s = z + S(x)\big(Q(z) - z - C\big), \tag{1}$$

with $z = |\nabla\Phi_N|/a_0$ and

$$S(x) = \frac{1}{1 + (x/x^*)^2}, \qquad x = \frac{c^2\|\nabla\nabla\Phi_N\|}{a_0^2}.$$

The interpolating function that multiplies baryonic acceleration does not contain C,

$$\nu_{\mathrm{eff}} = 1 + S(x)\big(\nu_{\mathrm{std}}(y) - 1\big). \tag{2}$$

C remains in the shell stress,

$$W_{ij} = -\frac{c^2}{2}\,S'(x)\,\big(P(y) - C\big)\,\frac{H_{ij}}{H_n}, \tag{3}$$

where $P(y) = \int_0^y 2t\,(\nu_{\mathrm{std}} - 1)\,dt$. For a spherical shell the outward piece is

$$g_W = \frac{2}{\sqrt{6}}\,r^{-3}\frac{d}{dr}(r^3\tilde{w}).$$

The frozen constants are $a_0 = 1.1607 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $x^* = 7.035884281332817 \times 10^{11}$. Neither is fitted in the tests below.

2 Non-relativistic action

The action has two potentials. Φ couples to matter. An auxiliary potential $\hat{\Phi}$ carries the invariant

$$x = \frac{c^2 \|\nabla\nabla\hat{\Phi}\|}{a_0^2},$$

which is computed from $\hat{\Phi}$ and is not varied. The action is

$$S = -\frac{1}{8\pi G}\int d^3x \Big[2\nabla\Phi\cdot\nabla\hat{\Phi} - a_0^2 Q(|\nabla\hat{\Phi}|^2/a_0^2) \\ - a_0^2\big(S(x)-1\big)\big(Q(|\nabla\hat{\Phi}|^2/a_0^2) - |\nabla\hat{\Phi}|^2/a_0^2 - C\big)\Big] - \int d^3x\,\rho\Phi, \tag{4}$$

with $Q' = \nu_{\rm std}$. Variation with respect to Φ gives $\nabla^2\hat{\Phi} = 4\pi G\rho$, so $\hat{\Phi} = \Phi_N$. Variation with respect to $\hat{\Phi}$, at fixed x, gives $\nabla\Phi = \nu_{\rm eff}\nabla\Phi_N$. Setting $S = 1$ recovers QUMOND. Setting $S = 0$ recovers Newton. The derivative of the action with respect to the constant C is proportional to $\int (S-1)\,dV$. On the solar shell that integral is $-2.06\times10^{46}\,\mathrm{m}^3$, which is not zero, so the field equation does not choose C. A boundary condition that the screened source vanish at zero field forces $C = 0$. That value fails the Oort test and is the subject of the companion paper. The working value is fixed by the Oort window, once, and is then held fixed.

With x not varied, the action has only first derivatives of the two potentials. The higher-derivative question appears when x is promoted to a dynamical field. That promotion is not carried out here. The shell on which S' would become that higher derivative peaks at $1.17\times10^4\,\mathrm{au}$, with half-maximum from 8.74×10^3 to $1.75\times10^4\,\mathrm{au}$, a fractional width 0.75. It is not a thin correction.

3 One field is not enough

A single scalar has one gradient at one point. Read as a Galactic external field at the Sun, Cassini needs that gradient, in units of a_0, between 1.637 and 2.068. Read as a galaxy in the field of M31, the Milky Way needs 3.0×10^{-3} at the same point. The ratio is about 550 to 690. The tide takes both values: $x \approx 1.8\times10^{12}$ at the Sun's MOND radius and $x \approx 3.9\times10^7$ in the disk at the solar circle, so $S \approx 0.133$ and $S \approx 1$.

4 The Oort window

For a uniform external field the binding offset of a comet that falls from rest is linear in C,

$$\frac{1}{a}(C) = \frac{1}{a}(0)\,\frac{P(y_e) - C}{P(y_e)}. \tag{5}$$

At $g_e = 1.90\times10^{-10}\,\mathrm{m\,s^{-2}}$, $P = 0.750345$ and $(1/a)(0) = 1.385818\times10^{-4}\,\mathrm{au^{-1}}$. The observed peak, 10^4 to $10^5\,\mathrm{au}$, sits at

$$C \in [0.208899,\ 0.696200],$$

with the centre $C = 0.579124707982253$ at 31,623 au. At $g_e = 2.32\times10^{-10}\,\mathrm{m\,s^{-2}}$ the window is $[0.384925,\ 0.872226]$ and the centre is 0.755150. Every test below uses the centre at 1.90×10^{-10}, except the wide-binary window scan, which uses all three.

A full-map integral, including the divergence of W, gives a mean offset $1.582166 \times 10^{-5}\,\mathrm{au}^{-1}$ at the working C. The weighted median original semimajor axis is 63,234 au. The torque jump is $a_J = 3.669\,\mathrm{kAU}$, the same value as at $C = 0$.

5 Five tests at the centre

Cassini. The action-form quadrupole, in units of $10^{-27}\,\mathrm{s}^{-2}$, is 5.200000 at $g_e = 1.9 \times 10^{-10}$ and 3.991075 at 2.4×10^{-10}, both at $C = 0$ and at the working C. Both sit inside $[-2.0,\ 5.2]$. A constant C integrates to $-2.3 \times 10^{-42}\,\mathrm{s}^{-2}$ against a $P(y)$ piece of 4.44×10^{-27}. The quadrupole is carried by the radial variation of P between 2.8×10^{3} and 1.25×10^{4} au.

Gaia wide binaries, centre. The clean cut, 14,542 pairs, at $g_e = 1.9 \times 10^{-10}$, gives raw $\chi^2 = 4.360$ and calibrated 2.995, against Newton 9.508 and an injection maximum of 20.431. The loose cut, 20,156 pairs, gives raw 39.605 and calibrated 18.171, under the bar of 18.5. The pre-registered exclusion does not fire. The 2.4×10^{-10} column was not rerun.

SPARC. One hundred seventy-one galaxies. The screen total is 18123.519. Newton in the same fitter is 293096.718. The unscreened μ_{std} total is 18123.521. The shell term does not move the aggregate. A disk solution of W, as opposed to the spherical estimate, is not in this number.

Four dwarfs. Crater II, Carina, Leo II, and Sculptor, at the measured dispersion minus its lower error. The decision mass-to-light ratios are 0.0599, 0.929, 2.659, and 1.335. All four are at or under 3. The Newtonian control matches the published Newtonian ratios to a relative difference of 3.2×10^{-7}. Leo II is not brought down by W. At the decision mass, $g_W = -8.0 \times 10^{-20}\,\mathrm{m\,s}^{-2}$. The ratio 2.659 is the Wolf formula at $\nu_{\mathrm{eff}} g_N$ with S near 1.

6 The window on the clean wide-binary cut

The same clean cut and the same bar were run at the two edges. A model fails only if the calibrated χ^2 exceeds 18.5 and the raw χ^2 exceeds that model's injection maximum. Newton on this cut is raw $\chi^2 = 9.508$.

C	raw χ^2	calibrated	κ	injection max
0.208899	2.248	1.446	1.555	22.315
0.579125	4.360	2.995	1.456	20.431
0.696200	5.186	3.878	1.337	20.387

None of the three fail. The loose cut was not repeated at the edges. The centre row is the earlier receipt. The edge rows are a separate run at the same x^* and the same g_e.

7 Radial comet orbits

Thirteen aphelia from 3×10^{3} to 3×10^{5} au were integrated in $-\mathrm{GM}/r^2 + g_W(r; C)$, with angular momentum aimed at a 5 au perihelion. The Newtonian osculating $1/a$ at perihelion matches the

potential to $1.26 \times 10^{-6}\,\mathrm{au}^{-1}$. Osculating semimajor axes for three starts inside or beside the observed window are:

start (au)	$C = 0$	0.209	0.579	0.696
13,925	turns at 9,683	4,204	5,771	6,542
30,000	4,877	6,006	10,192	13,073
94,868	6,270	8,271	19,032	32,342

At $C = 0$, two starts in the shell never reach the inner system, and the ones that do land below 10^4 au. At the centre, a 3×10^4 au start lands at 1.02×10^4 au and a 9.5×10^4 au start lands at 1.90×10^4 au. At the high edge the upper half of the window stays inside 10^4 to 10^5 au. This is a radial integral. It is not a population integration with stellar encounters and the Galactic tide of the kind run by Vokrouhlický, Nesvorný and Tremaine. That population remains open.

8 What is not claimed

There is no cosmological sector in this action. The branch in which x is a dynamical relativistic field is open. The constant C is fixed by the Oort window, not derived from a symmetry. The $C = 0$ failure is the companion paper. An intermediate wide-binary mask between the clean and loose cuts is not a nested sample, and no model χ^2 was computed on it. x^* is the Cassini value, not a derived constant.